Project Protocol:

Bank Marketing Offering

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Project Background

In the banking industry, marketing teams often face the challenge of deciding **which customers are most likely to accept a new credit card offer**. Sending offers to every customer is expensive, inefficient, and risks customer fatigue. On the other hand, missing out on high-potential customers means lost revenue opportunities.

To simulate this real-world challenge, I created a synthetic dataset that mirrors the type of information a bank's marketing department might collect:

- **Customer demographics** (age, income, etc.)
- **Transaction behavior** (spending activity, frequency)
- **Engagement history** (previous offers sent and accepted)
- **Derived features** (e.g., acceptance ratios over time)

The goal of this project is to build and evaluate machine learning models that can predict whether a customer will accept a targeted credit card offer. While the data is simulated (to avoid confidentiality issues), the process closely reflects the decision-making steps that a real marketing team would take to optimize targeting strategies.

Why This Matters

Accurate prediction of offer acceptance has a direct business impact:

- **Reduced Costs** → Fewer wasted offers sent to customers unlikely to accept, saving on marketing spend.
- **Higher Conversion Rates** → By focusing on customers most likely to say yes, the campaign becomes more profitable.
- **Improved Customer Experience** → Customers receive fewer irrelevant offers, which helps build trust and long-term loyalty.
- **Smarter Resource Allocation** → Marketing teams can test multiple strategies on high-potential segments before scaling campaigns.

Although the dataset in this project is synthetic, the insights and methods mirror the **real-world decision-making process inside a bank's marketing department**.

Data source

- **Synthetic dataset** generated programmatically by ChatGPT to mirror real retail-bank marketing data while protecting confidentiality.
- Distributions were chosen to reflect realistic patterns: **skewed spending**, **class imbalance** in the target (acceptance is rarer), **mixed data types** (numeric, categorical, booleans), and **structured missingness**—all typical in banking campaigns.

Shape

- **Rows:** 25,795 customers
- **Columns used for modeling:** 20 features + 1 binary target (offer_accepted)
    - (After encoding and feature engineering, we train on 20 predictors; the original text categorical columns were encoded and then dropped.)

```
 #   Column
---  ------
 0   customer_id
 1   age
 2   gender
 3   marital_status
 4   income_bracket
 5   education_level
 6   region
 7   num_transactions_last_6_months
 8   avg_transaction_value
 9   credit_score
 10  online_activity_score
 11  has_mobile_app
 12  avg_login_frequency
 13  num_complaints_last_year
 14  last_offer_accepted
 15  num_past_offers_sent
 16  preferred_contact_channel
 17  last_contact_response_time_days
 18  offer_accepted
```

1. Data Preparation:

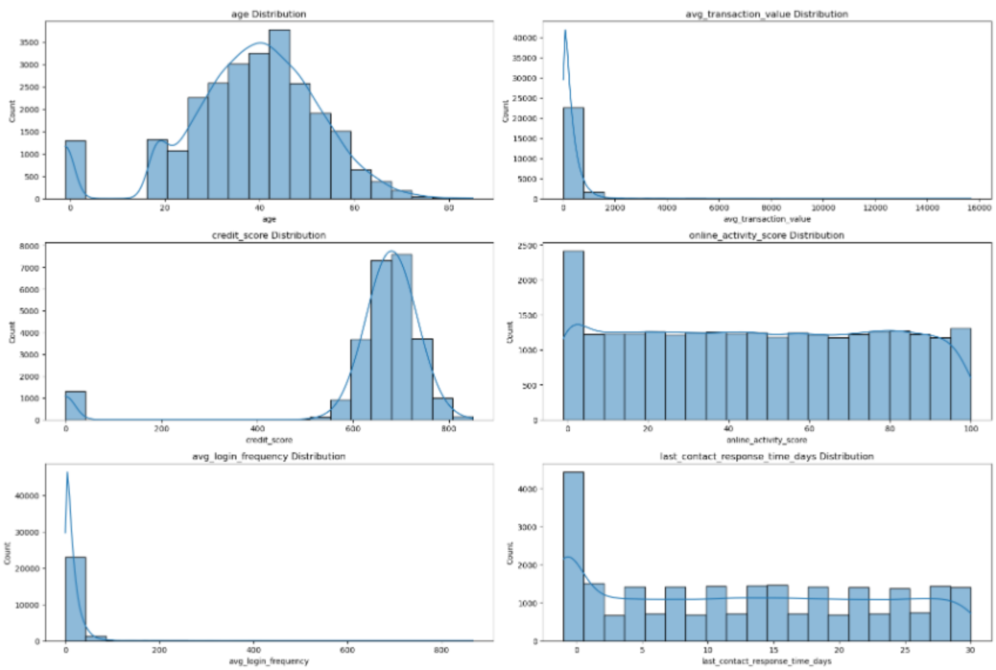
- **Dropped irrelevant columns:** removed customer_ID and technical fields not useful for modeling.
- **Corrected data types:** converted object-type features to their true types (int, float, or category).
- **Missing values check:** identified null counts per column but did not impute them yet (reserved for later step).
- **Categorical feature review:** inspected categories for imbalance, rare levels, and potential encoding strategies.

2. Exploratory Data Analysis (EDA):

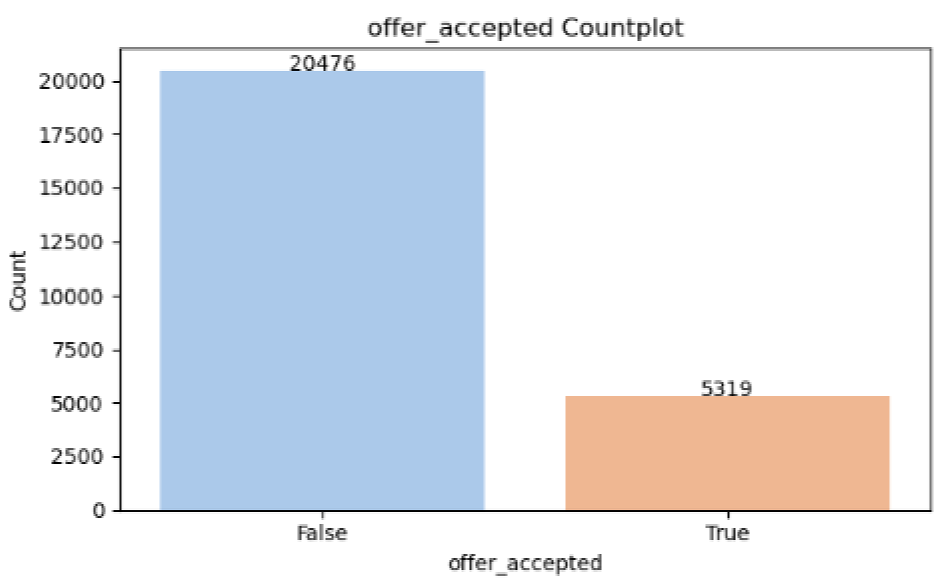
2.1 Feature distributions

Visualized numeric, categorical, and dummy features (which is just our target feature) to understand their spread and balance.

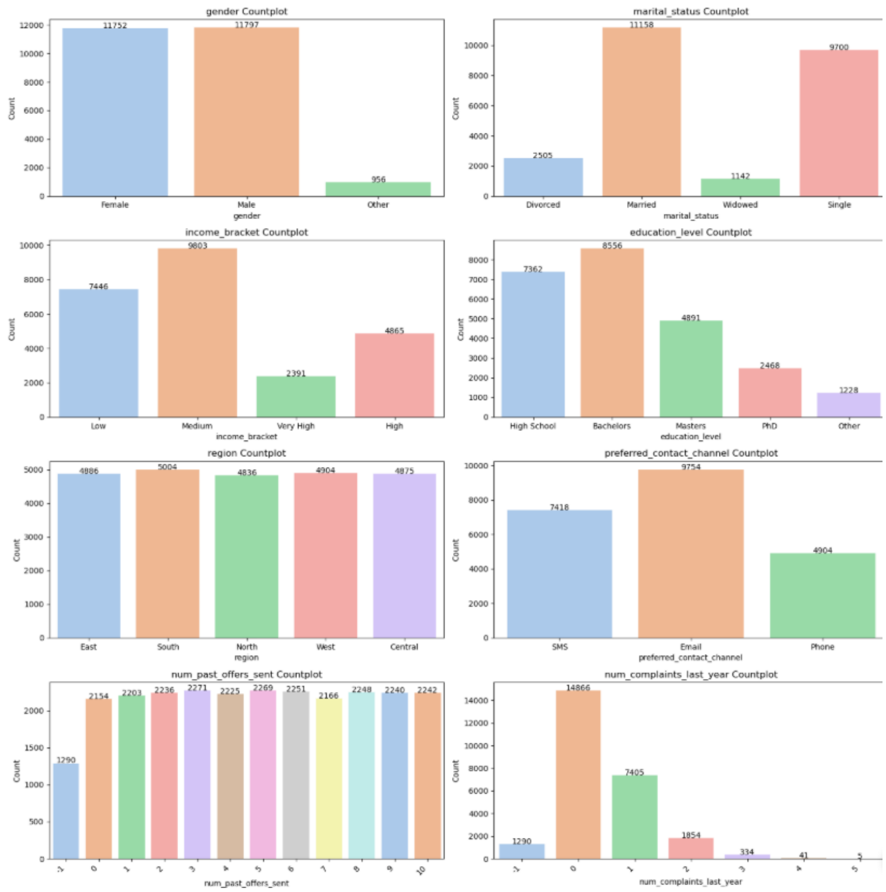
Numeric:



Target:



Categorical:



2.2 Skewness check

Measured skewness of numerical features to evaluate potential transformations.

```
Skewness Analysis:

age - Skewness: -0.64 => Moderately skewed
avg_transaction_value - Skewness: 10.53 => Highly skewed
credit_score - Skewness: -3.48 => Highly skewed
online_activity_score - Skewness: 0.02 => Approximately normal
avg_login_frequency - Skewness: 11.72 => Highly skewed
last_contact_response_time_days - Skewness: 0.12 => Approximately normal

Summary:
Total Normal Features: 2
Total Non-Normal Features: 4
```

Business Insights from the Distributions

- **Customer Spending (avg_transaction_value)** is heavily skewed — a small segment of users spend much more than average.
  🔸 *Implication:* High-spending customers may represent a lucrative but niche target for premium offers.

- **Credit Score (credit_score)** skew suggests many customers have very low or very high scores, with fewer in the mid-range.
  🔸 *Implication:* Marketing strategies may need to segment users into "credit-challenged" vs. "credit-eligible" groups.

- **Login Frequency (avg_login_frequency)** shows a heavy right skew, indicating some highly engaged users, but most log in infrequently.
  🔸 *Implication:* Campaigns may need to re-engage inactive users or reward highly active ones to boost overall engagement.

- **Online Activity Score (online_activity_score)** and **Response Time (last_contact_response_time_days)** appear balanced.
  🔸 *Implication:* These features are likely reliable and consistent across the customer base, making them valuable for modeling and segmentation without major preprocessing.

2.3 Correlation analysis

Given that most numeric features in the dataset are non-normally distributed, we applied Spearman's rank correlation to assess monotonic relationships between them.

Key Findings:

- All correlation values fall in the **very weak** range (between -0.01 and +0.01).
- No feature pair shows a meaningful or strong monotonic relationship.
- This suggests that each feature captures a distinct aspect of customer behavior or profile.

Business Insight:

Despite high skewness in features like avg_transaction_value and credit_score, their lack of correlation with other features suggests **non-redundancy**. Each variable may provide **unique value** in predicting the target — an encouraging sign for modeling.

In plain terms, customer age, behavior, credit data, and response patterns appear to act **independently**, which means a well-regularized model might benefit from keeping them all.

2.4 Association tests for categorical features

Chi-square test for categorical features.

Feature Pair	p-value	Strength of Evidence	Key Insight
`gender` × `marital_status`	0.0929	Weak evidence	Mild indication that gender and marital status may be related
`gender` × `preferred_contact_channel`	0.0986	Weak evidence	Possible relationship — worth exploring via visualization or modeling
`num_past_offers_sent` × `offer_accepted`	0.0935	Weak/moderate evidence	Slight tendency: more past offers might influence acceptance rates
`preferred_contact_channel` × `offer_accepted`	0.1501	Very weak trend	Weak signal — not statistically significant, but relevant for business logic
`region` × `preferred_contact_channel`	0.1590	Very weak trend	Slight regional pattern in contact preference — minor, but might guide marketing strategies
Most other pairs	> 0.20	No evidence of association	No detectable statistical relationship

Key Insights: No strong associations were detected between categorical features and the target (offer_accepted).

A few weak signals may be worth tracking, especially:

Preferred contact method (as it intersects with both region and gender)

Historical engagement (num_past_offers_sent)

There is no multicollinearity concern among categorical features based on Chi-Square.

These results suggest that categorical features alone may not be strong predictors, but can still provide business context or interactions with numeric features.

Cramer's V calculated to measure strength of association.

Key Takeaways:

- **All categorical pairs show very weak associations**, including those involving the target offer_accepted.
- This implies that **categorical features individually have minimal predictive power** based on their mutual associations.
- The very low Cramér's V values support previous chi-square results that showed no significant dependency.
- Suggests that the model will likely rely more on numeric features or on complex interactions rather than on simple categorical relationships.

2.5 Numeric features vs. target

ANOVA test used to evaluate which numeric features significantly differentiate between target classes.

Statistically Significant Differences ($p < 0.05$)

Feature	Insight
`avg_transaction_value`	Customers who accepted the offer had significantly different average transaction values compared to those who didn't.
`credit_score`	Credit score distributions vary significantly between the two groups. Higher scores may correlate with acceptance.
`online_activity_score`	Online engagement levels are significantly different across acceptance groups. Possibly a behavioral predictor.

No Significant Differences ($p \geq 0.05$)

Feature	Insight
`age`	No statistically meaningful difference in average age between the groups.
`avg_login_frequency`	Login behavior does not significantly distinguish between responders and non-responders.
`last_contact_response_time_days`	Response delay to last contact does not show a significant difference across groups.

2.6 Outlier treatment

Inspected numeric variables with **boxplots**, and applied **IQR method** to treat outliers.

	feature	outliers_cnt	distribution_changed	mean_diff	drop
0	age	1325	+	NA	no
1	avg_transaction_value	1271	+	NA	no
2	credit_score	1404	+	NA	no
3	online_activity_score	0	-	NA	no
4	avg_login_frequency	1321	+	NA	no
5	last_contact_response_time_days	0	-	NA	no

Business Interpretation:

While several features show a substantial number of outliers and a statistically significant change in distribution after removing them, the mean differences across target groups ('offer accepted' or not) do not indicate meaningful shifts (marked as 'Not Applicable' due to unavailable mean difference data). This suggests that although outliers influence the shape of the data, they do not significantly affect the relationship between these features and the target variable.

Decision: We will **retain all outliers** in the dataset. Removing them could distort the natural variability in customer behavior without providing meaningful improvement in model performance or interpretability.

This approach helps preserve the complexity and real-world nuances embedded in the synthetic dataset, ensuring our predictive models are robust to typical customer data irregularities.

2.7 Missing value imputation

After EDA insights, imputed missing values using strategies aligned with feature type (mean/median for numeric, mode or category for categorical).

Missing Data Imputation

To handle missing values in our dataset, we applied different imputation strategies based on feature types:

- **Numeric features** (e.g., avg_transaction_value, avg_login_frequency):
  We used **median imputation** because the median is robust to outliers and skewed distributions, which are common in financial and behavioral data. This approach replaces missing values with the median of the observed values, preserving the central tendency without being influenced by extreme values.
- **Categorical features** (e.g., gender, marital_status, income_bracket):
  We applied **mode (most frequent category) imputation**. This method replaces missing values with the most common category in each feature. It preserves the existing category distribution and is appropriate when missingness is assumed to be random or non-informative. Alternatively, a "Missing" category can be introduced if missingness is believed to carry meaningful information.

This targeted approach ensures that each feature type is treated appropriately, maintaining data integrity and preparing the dataset for accurate model training.

3. Feature Engineering

After preparing the data, we focused on enhancing the dataset with meaningful transformations and new variables.

1. **Encoding categorical features**
   - Applied label encoding / one-hot encoding to convert categorical features into numerical form suitable for modeling**.**
2. **Feature creation**
   To capture deeper customer behaviors and improve model predictive power, we engineered several domain-inspired features:
   - **avg_transaction_value_per_income_bracket**
     Normalizes spending across different income levels by dividing

transaction value by income bracket. Helps compare customers with different earning capacities.

  - **spending_per_login**
    Captures how much customers spend per login session. Defined as average transaction value divided by average login frequency. Indicates profitability of digital engagement.

  - **complaints_per_transaction**
    Measures dissatisfaction by calculating the number of complaints relative to the number of transactions in the last 6 months. Useful for detecting churn risk.

  - **offers_response_ratio**
    A key predictor in this dataset. Calculates the ratio of accepted offers to past offers sent, reflecting customer responsiveness to marketing campaigns.

  - **days_since_last_contact_normalized**
    Adjusts recency of last contact by dividing days since last contact by customer's age, providing lifecycle-aware engagement insights.

4. Feature Selection:

To ensure our models train on the most informative predictors, we evaluated features using five different methods:

- **Lasso (Logistic Regression with L1 penalty)**
- **Ridge Classifier**
- **Gradient Boosting Classifier**
- **Random Forest Classifier**
- **XGBoost Classifier**

Each method produced a binary vote per feature (1 = important, 0 = not important). We summed the votes to measure overall consensus.

Key Observations:

- Features with a **perfect score of 5** were unanimously selected by all models, showing strong predictive potential.
- Features scoring **4** were selected by most models except one (Gradient Boosting in this case).

Rationale for keeping features scoring 4:

- A score of 4/5 still indicates high consensus on feature importance.
- Gradient Boosting may have unique sensitivities; excluding features solely due to one model's vote could discard useful information.
- Retaining these features allows us to validate their true impact during model training and in interpretability analyses (e.g., feature importance or SHAP values).

This balanced approach prevents premature feature exclusion and keeps our feature space robust for modeling.

5. Model Selection:

We evaluated multiple classification models to predict customer offer acceptance. Our evaluation metrics included **Balanced Accuracy** (to ensure fair performance across both classes) and **Matthews Correlation Coefficient (MCC)** (to capture overall predictive quality even with class imbalance).

	Model	TP	FP	TN	FN	Accuracy	Precision	Recall	F1 Score	Balanced Accuracy	MCC
0	XGBoost	968	23	4101	67	0.982555	0.976791	0.935266	0.955577	0.964844	0.945075
1	Gradient Boosting	958	2	4122	77	0.984687	0.997917	0.925604	0.960401	0.962559	0.951960
2	Random Forest	952	0	4124	83	0.983912	1.000000	0.919807	0.958228	0.959903	0.949558
3	Logistic Regression	790	101	4023	245	0.932933	0.886644	0.763285	0.820353	0.869397	0.782712
4	Ridge Classifier	758	388	3736	277	0.871099	0.661431	0.732367	0.695094	0.819142	0.614917
5	SVC	674	1407	2717	361	0.657298	0.323883	0.651208	0.432606	0.655017	0.253086
6	KNN	182	330	3794	853	0.770692	0.355469	0.175845	0.235294	0.547913	0.128349

Key Insights Prior to Fine-Tuning:

- Gradient Boosting, XGBoost, and Random Forest were the top three models, showing high and consistent performance across balanced accuracy and MCC.

- These models were selected for the fine-tuning stage to optimize hyperparameters and determine which generalizes best.

Model Fine-Tuning Results

After hyperparameter optimization using GridSearchCV, we compared the top three models based on **balanced accuracy**:

Model	Best Parameters	Best Balanced Accuracy
Gradient Boosting	{'learning_rate': 0.2, 'max_depth': 3, 'n_estimators': 100}	0.9618
XGBoost	{'colsample_bytree': 0.8, 'learning_rate': 0.1, 'max_depth': 3, 'n_estimators': 100}	0.9624
Random Forest	{'max_depth': None, 'max_features': 'sqrt', 'min_samples_leaf': 1, 'min_samples_split': 2, 'n_estimators': 100}	0.9587

Interpretation:

- XGBoost slightly outperformed the others with the highest balanced accuracy, followed closely by Gradient Boosting and Random Forest.
- All three models are strong candidates for the final model selection step.

Feature Importance

We performed feature importance analysis using the final XGBoost model.

Key Observation:

- offers_response_ratio, an engineered feature, has the highest influence on model predictions, demonstrating the value of feature engineering.
- Other features, while individually less influential, still contribute to overall predictive performance.

Insert feature importance plot screenshot here

Displaying feature importance helps maintain transparency and provides insight into how different features affect the model's decisions.

Next Steps

- **Final Model Validation:** Evaluate the selected XGBoost model on an external validation set or through cross-validation to confirm generalization.
- **Deployment Considerations:** Integrate the model into marketing campaign decision pipelines, providing actionable insights for targeting offers.
- **Continuous Improvement:**
    - Incorporate new customer behavioral data over time to retrain and update the model.
    - Explore additional feature engineering opportunities, including interaction terms or temporal features.

Thank You!